FINQUEST INDIA

M2 - Mid-submission Progress Report | Megathon X: Singularity

Project	Interactive financial-literacy platform for Grade 11-12 students
Current focus	Bilingual learning journeys, practical money labs, calculation checks, and safe simulations
Submission stage	M2 progress snapshot - 11 October 2026

1. Current Approach and Changes Since M1

M1 proposed a broad financial learning and planning platform for ages 12 to 60+. The current repository narrows the first build to Grade 11-12 learners, including students in underserved communities around Hyderabad. FinQuest turns key money concepts into short lessons, guided decisions, interactive labs, knowledge checks, and progress evidence. It is educational only: simulations use hypothetical assumptions and do not execute real-money transactions or recommend products.

Current prototype approach

PLAN & BUDGET	Needs versus wants, spending trade-offs, budget allocation, and unexpected expenses.
SAVE & MANAGE	Simple/compound interest, fixed deposits, recurring deposits, inflation, and purchasing power.
UNDERSTAND INVESTING	Shares and market mechanics, mutual funds, NAV, fund types, fees, and SIP illustrations.
STAY FINANCIALLY SAFE	Phishing, UPI/OTP safety, scam red flags, and routes to trusted reporting resources.

Implemented structure: the React application exposes Home, Learn/pillar/lesson, Money Lab, Explore/Glossary, Practice, Progress, Dashboard, and Investing Made Simple routes. The curriculum registry contains 16 lessons, with English and Telugu stage content, learning objectives, a concept visual, an interaction, and at least three mastery-check questions per lesson. The glossary has 45+ bilingual terms. Lesson stage/visit/objective evidence is stored in browser local storage; quiz selections are not restored after reload.

Key refinements since M1: moved from a lifelong general concept to a school-focused build; organized content around four challenge pillars; added interactive finance and safety labs with formula-level tests; made bilingual curriculum checks explicit; and documented remaining browser, accessibility, localization, and performance gaps. Some legacy simulator controls and explanatory copy are still English-only.

2. Experiments Run So Far and Results

The repository audit dated 11 October 2026 records code-level verification. These results establish implementation checks, not student learning outcomes; no formal classroom user study or pre/post learning measurement is recorded.

Check	Evidence and recorded result
Automated tests	55 tests passed across 5 files using **npm test -- --pool=threads --maxWorkers=1**, as recorded in the curriculum audit. Coverage includes curriculum registry, lesson-route smoke checks, and financial formula cases.
Production build	Passed in the audit using **npm run build** (TypeScript plus Vite build). A large main JavaScript chunk, about 1.20 MB before gzip and about 332 KB gzip, remains a performance/code-splitting opportunity.
Financial calculations	Test cases cover simple/compound interest, quarterly FD maturity, inflation-adjusted purchasing power, both SIP contribution timings, NAV, market-cap/earnings ratios, CAGR, real return, Rule of 72, budget balance, and invalid/boundary inputs.
Curriculum integrity	Automated checks cover four pillars, valid prerequisites, bilingual lesson text and glossary, lesson objectives, visual/interaction presence, and at least three mastery questions per registered lesson.

Reproducibility note: findings above are taken from the public repository audit and test-source documentation; this report does not claim that the web app was manually verified in a live browser during preparation.

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Validation status, remaining work, and risks

3. Progress on Testing

Status	Current evidence / limitation
RECORDED AS PASSED	Automated test suite and production build are reported as passing in **CURRICULUM_AUDIT.md**. Formula tests include normal cases and selected zero, negative, and non-finite inputs.
NOT YET VERIFIED	The audit records attempted browser checks that timed out. Route-by-route visual behavior, console errors, mobile/tablet layouts, zoom, keyboard-only navigation, screen-reader labels, and end-to-end flows therefore remain unverified.
NOT YET MEASURED	No controlled student task-completion study, quiz-score improvement, confidence pre/post measure, or teacher/facilitator feedback has been collected.

4. Plan for the Remaining Time

Priority	Action	Evidence to capture
1. Core flows	Run browser end-to-end checks for budget updates/deficit, FD and SIP calculations, NAV/unit arithmetic where available, quiz retry/completion, progress persistence, reset, and invalid inputs.	Dated checklist with pass/fail outcomes, test inputs, expected values, screenshots, and defect IDs.
2. Browser UX	Verify main routes in a real browser at phone, tablet, and desktop widths; check keyboard flow, focus visibility, readable labels, zoom, contrast, and console/network errors.	Viewport screenshots, keyboard checklist, and resolved issue log.
3. Content and data	Confirm assumptions and source/date labels wherever current financial information is shown; test missing/stale-source fallbacks; keep hypothetical scenarios visually separate from real data.	Source/date snapshots and failure-state evidence; no invented values.
4. Learning quality	Localize remaining English-only controls/copy (including legacy SIP, indices, company financials, ratios, and RD interactions); improve recap and misconception feedback.	Updated bilingual screens and a short change log.
5. Release readiness	Revisit bundle-size/code-splitting opportunity, then invite a small intended-learner group or facilitator to try a budget mission, explain NAV, and complete a safety scenario with appropriate consent and safeguards.	Build/test log, learner task observations, anonymized feedback, and confirmation that changes are pushed to main.

5. Risks, Mitigations, and Current Outcome

Data freshness: external sources may be unavailable or delayed; show source and last-updated date and explain fallback states. Learning versus gaming: provide reasons and reflection, not points alone, and never imply guaranteed returns. Privacy and minors: retain the local-storage disclosure, avoid unnecessary personal/financial data, and obtain appropriate guardian consent for learner testing.
Current outcome: FinQuest India has a structured, bilingual curriculum and a code-tested interactive prototype. The repository audit records passing automated checks and a production build, but browser-level usability/accessibility validation and classroom impact remain open. No real-money transactions or personalized investment recommendations are in scope.

Repository basis: github.com/umamaheswarigww-cyber/Vanguard (main); README.md, CURRICULUM_AUDIT.md (audit dated 11 October 2026), src/App.tsx, src/lib/finance/formulas.ts, src/tests/finance.test.ts, and src/tests/curriculum.test.ts.

Prototype disclaimer: all calculator outputs are illustrative and depend on stated assumptions. This website is for financial education, not financial advice.